NRG PLATFORM · PHASE 1

Feature status

What the platform already does, what is still outstanding, and what is holding each outstanding piece up.

Updated 9 November 2026 · Prepared for Ian Forde and Jade Nicome · Kept up to date as work lands

How to read this

Every feature carries one of four labels. "Staging" is the test site (nrg-platform-staging.vercel.app); "prod" is the live site students use.

LIVE	On the live site for students now.
STAGING	Built and tested on the test site; waiting for a review, a decision or a go-live step before it reaches students.
PARTLY BUILT	Some of it works; the rest is listed in Part B.
NOT BUILT	Not started. Part B says what it is waiting for.

12	11	4	7
feature areas live	built, on staging	partly built	not built yet

The single biggest gate is content review: most of what is "on staging" is waiting for Jade to approve questions, cases or reviews, or for Ian to switch something on for the live site. Part B lists every blocker with who can clear it.

Part A. What is already built

Grouped by who uses it. Each entry says what it does in plain terms and where it is.

For students: studying

Practice questions (tutor mode)

LIVE

Students pick a domain, topic, taxonomy level or difficulty and answer questions one at a time. After "Check answer" they see the correct option, an explanation and a rationale for every option, colour-coded as correct, close, priority or incorrect. Answer keys never reach the browser before the student answers.

Keyboard shortcuts (1–9 or A–D to choose, Enter to check and move on) are on staging since 9 Nov.

Flashcards

LIVE

Rapid recall by topic: flip, shuffle and drill.

Mock exams

LIVE

Exam-format papers set by a teacher, taken alone or in a group of up to five, with an optional timer. No feedback while answering; explanations stay hidden until the teacher releases them. Each result includes a fatigue analysis: accuracy and pace by quarter of the exam, and a recommendation when accuracy drops late.

Topic reviews (15)

STAGING

Jade's 15 slide decks turned into readable reviews (normal function, what goes wrong, assessment, management, care plan, how the exam tests it), each with a practice question and a "practise this topic" button. Questions link back to the matching review after they are answered. Reading a review to the end earns 30 XP. The library has a filter panel (search, body system, read or not, sort, card or list view).

All 15 are still drafts: students see none until they are published (Part B).

Case studies

PARTLY BUILT

An unfolding-case player: the patient's story moves through stages with vitals, labs, medications, a calculator and reference ranges; answers lock per stage and feedback comes at the end. Teachers review, validate and publish cases; a generator drafts new ones to Jade's standard.

The player is live; 6 pilot cases wait for Jade's validation, so students have no cases yet.

My progress (private analytics)

STAGING

New on 9 Nov (decision D8, option 2). Each student sees only their own figures, never compared with classmates:

- Rank, questions answered, accuracy, days studied, domains practised.
- Strengths and weaknesses by RENR domain (New / Weak / Developing / Mastered, using the ranking rules) and by taxonomy level, each with a "practise" link.
- Weakest and strongest topics, linked to the matching topic review and to practice on that topic, plus a "focus next" card.
- Exam stamina from their mock exams, and a 12-week study calendar.

Mock answers count only once the teacher releases that exam's explanations, so the page can't give away which answers were right.

Search and shortcuts

STAGING

Ctrl+K (⌘K on a Mac) or "/" opens a search box that jumps to any study page, topic review or domain practice.

Nursing Knowledge facts (133)

PARTLY BUILT

A rotating "did you know" block of short clinical facts. Built; waiting for Jade's clinical sign-off before students rely on it.

For students: motivation and rank

XP, ranks and study streak

LIVE

Students earn XP for practice, mock exams and approved questions, and climb Jade's 25-level Benner ladder (Novice to Expert). A daily study streak counts Trinidad days.

Ranking v2 and promotion quizzes

STAGING

Weighted XP (more for weak topics, less for mastered ones), a streak bonus, a mastery milestone, a daily cap, and promotion by a 25-question timed quiz instead of automatic level-ups.

Running in "shadow" on the live site: it calculates quietly next to today's rules so the two can be compared before launch.

Trophies and celebrations

STAGING

56 badges (a rank badge per promotion, tier trophies, and accomplishments such as first mock, streaks and mastered domains), a rank-up ceremony, a trophy case, and a classmates' wall students can opt out of. Launches with ranking v2.

Write a question + AI grading

STAGING

Students write their own question on a topic the platform assigns; an AI grader scores it against Jade's template with feedback per criterion, and teachers see the grade. Writing questions has been live since September; the AI grading, the optional 24-hour rule and the writers' leaderboard are on staging, switched off until Jade chooses (D1–D7).

For students: look and feel

Redesign and four themes

LIVE

The whole app restyled; students choose Royal, Teal, Midnight (dark) or Sunset, each with its own fonts. Switching is instant (a lag fix is on staging since 7 Nov).

Profile, messages and calendar

LIVE

Profile picture and name, private messages with teachers (a student can cite a mock attempt), and an event calendar.

For teachers and admins

Question review queue

LIVE

Teachers approve, edit or send back questions; every AI question needs a named reviewer before it can go live.

Generation requests

LIVE

Jade asks for new questions or cases either by picking domain, topic, taxonomy and difficulty, or by describing what he wants in his own words; the request is queued for the AI author and the results land in his review queue.

Mock exam builder and release

LIVE

Build question sets, set a timer, assign to students or groups, and release explanations when ready. A cohort fatigue card shows whether the class tires late in exams.

Class analytics (at-risk list and class trends)

STAGING

A teacher page listing students at risk and why: no answers for 7 days, under 50% over their last 50 answers, a recent mock below the pass mark, or a 15-point accuracy drop in a fortnight. Each reason shows the student's own numbers, with a one-click message to that student. Below it: 12 weeks of students practising, class accuracy and average mock score, and class accuracy by domain against the month before. The teacher dashboard shows how many students are at risk. Students never see this page.

Thresholds are settings, adjustable without code. Three are proposals for Jade: the mock pass mark (70%), how long a mock counts as recent (30 days), and the 20 answers needed before accuracy counts.

Accounts and access

LIVE

Email sign-up and verification, four roles (student, teacher, admin, super admin), teacher approval, suspending accounts. Admins can set a student's plan by hand.

Content protection

LIVE

A faint watermark with the student's identity on question pages, no copy or print, and blurring when the window loses focus or screenshot keys are pressed. Staff can reveal a watermark from a leaked screenshot. (No website can fully block screenshots; the watermark makes leaks traceable.)

Free plan limits

STAGING

Free students get 10 practice questions a day, no mock exams, and one topic review a day for 5 minutes. Enforced by the database. Built with the switch off; nothing changes for students until it is turned on.

Paid plans by bank transfer

STAGING

Free, 1 month (TT$200) and 3 months (TT$500). Students pay into NRG's account, upload the receipt and see its status; an admin matches it to the bank statement and approves or rejects with a reason. Receipts are checked automatically for AI-made images and reuse. Paid access ends by itself at expiry, and renewing early adds time on. Card payment shows as "coming soon".

Waits for NRG's bank details before students can pay.

One device per student, and shared-account alerts

STAGING

A student account can be signed in on one device at a time: a new sign-in signs the old device out and tells it why. Admins see a list of accounts that keep signing each other out or use many browsers, review them, and adjust the thresholds. Staff are not affected.

Recordings library

STAGING

Past recordings and lectures, like a video journal: filter by type, domain and length; a watch page with chapters, notes, a searchable transcript and captions; it remembers where you stopped and links to the topic review and practice. Paid plans watch everything; free students get samples. Teachers add recordings with a form. Two sample videos are up for review.

Security underneath

LIVE

Row-level security on every table, answer keys hidden from student accounts, and table permissions tightened to only what each role uses; automated tests check all of it.

Part B. What is outstanding and what is blocking it

"Who" is the person who can clear the blocker. Most items need a decision or a content check, not more building.

Feature	Status	What is blocking it	Who
Built, waiting to reach students			
Topic reviews (15)	STAGING	Each review needs a read-through and the "Publish" button. Endocrine still shows glucose in mmol/L, which the platform rule forbids; two small deck quirks to fix.	**Ian + Jade**
AI question bank (1,377 in the queue)	STAGING	Clinical review and approval of the rewritten AI questions, including 71 where the correct answer changed. The copy to the live site is ready once approved.	**Jade**
Case studies (6 pilots)	PARTLY	Validation of each case (its quality report lists what to confirm), and whether to keep the reference-range panel.	**Jade**
Ranking v2 + trophies on the live site	STAGING	1–2 weeks of shadow comparison, then an announcement (everyone restarts at level 0) before switching on. Jade to confirm four ranking settings.	**Ian, Jade**
Free plan limits on the live site	STAGING	Confirm the numbers (10 a day, no mocks) and whether promotion quizzes and flashcards stay free; choose when to switch on. Without a checkout, free students can only be upgraded by an admin.	**Ian + Jade**
AI grading of student questions	STAGING	Jade chooses instant (paid API) or queued (subscription) grading and the rules in D1–D7 of the decisions PDF.	**Jade**
My progress, search, review filters, keyboard shortcuts, theme fix	STAGING	Ian's review on staging; then the 9 Nov database change goes to the live site before the next release. Ian to confirm that practice may record each answer on the live site (XP unchanged).	**Ian**
Class analytics (at-risk list, class trends)	STAGING	Built on staging (9 Nov) to the at-risk definition Ian approved. Jade to confirm the mock pass mark (70% proposed; mocks had none), 30 days as "recent" for a mock, and 20 answers before accuracy counts. Then the database change goes to the live site.	**Jade, Ian**
Nursing Knowledge facts	PARTLY	Clinical sign-off on the 133 facts.	**Jade**
Live-site database changes	STAGING	Ten database changes (22 Oct and 9 Nov) must be applied to the live database before the next release. Ian runs them.	**Ian**
Partly built			
Mock exams to the v2 rules	PARTLY	Always 100 questions in 150 minutes, blueprint balance by domain and taxonomy, no going back, two 5-minute breaks. Should every mock be forced into this shape?	**Ian + Jade**

Feature	Status	What is blocking it	Who
Login security	PARTLY	2FA (by email code only, since there is no SMS provider?), 15-minute idle logout, lockout after failed logins, a school/university field (needs a privacy line), and custom email sending before real sign-ups.	**Ian**
Not built yet			
More topic reviews (Template 9 lists 50)	NOT BUILT	Content: 15 of 50 decks exist. Also whether writing a question after each review is required (D6).	**Jade**
Paid plans by bank transfer	STAGING	Built on staging (9 Nov): plans page (Free, 1 month TT$200, 3 months TT$500), payment form with receipt upload, automatic receipt check, admin approval page, plan status page; access ends by itself at expiry. To go live: NRG's bank details (D30), the database changes on the live site, and the email settings on the live site (emails and reminders are built and use Gmail on staging).	**Ian, Jade**
Account sharing guardrails	PARTLY BUILT	Built on staging (9 Nov): one signed-in device per student, and the admin list of accounts that look shared (5 sign-outs in a day or 4 browsers in a week, adjustable). Still to build: an email code on a new device, which needs custom email sending.	**Ian**
Recordings library (past recordings and lectures)	STAGING	Library and watch page built on staging (9 Nov) with two sample videos (ABG interpretation, patient demographics): filters, chapters, notes, searchable transcript, captions, resume, moving watermark, paid plans with free samples, and a teacher form to add recordings. Waits for Jade's review of the samples and his recordings, and a video host for long lectures (Ian).	**Jade, Ian**
Card payment online	COMING SOON	Shown on the plans page as "coming soon". Waits for a processor choice (WiPay, PayPal or other) and how its payouts reach the business account; no longer blocks selling plans.	**Ian + Jade**
Live study lobby, leaderboards, Google Meet debriefs	NOT BUILT	A real-time build; leaderboards conflict with keeping student comparison private. Needs a go-ahead and a decision on leaderboards.	**Ian + Jade**
Adaptive difficulty, personal study plan	NOT BUILT	Needs a few weeks of per-answer data (now being recorded); the class analytics it builds on is done.	**Later**
Evidence-base research agent	NOT BUILT	Low priority in the brief; needs a go-ahead.	**Ian**
Topic animations	NOT BUILT	Not specified: which topics, and what style.	**Jade**
Links from mock results to reviews	NOT BUILT	"Improve this domain" links on mock results and click tracking; small, can be built now on request.	**Ian**
Abbreviations, I/II/III and dose-calculation items	POLICY	Content rules not yet set: the abbreviation exempt list, whether the combination format stays, and how numeric dose items are handled.	**Ian + Jade**

Quickest unblockers

Jade: answer the decisions PDF (31 items, starting with D30: the bank details for paid plans), confirm the mock pass mark for class analytics, validate the 6 cases, and work through the AI question queue. **Ian + Jade:** publish the 15 topic reviews. **Ian:** apply the pending live-site database changes and decide on 2FA and mock exam rules.

Change log

9 Nov 2026	Class analytics built on staging: at-risk list with reasons, 12-week class trends, accuracy by domain.
9 Nov 2026	Recordings library built on staging with two sample videos.
9 Nov 2026	Recordings library planned (build plan written).
9 Nov 2026	Payment emails, plan reminders and receipt cleanup built; staging sends through Gmail for now.
9 Nov 2026	Paid plans by bank transfer built on staging (TT$200 / TT$500); shared-account alerts built.
9 Nov 2026	One device per student built on staging; prices in TT$.
9 Nov 2026	Plans set to Free, 1 month and 3 months; account sharing guardrails planned.
9 Nov 2026	Paid plans: bank-transfer route decided and planned; card payment marked coming soon.
9 Nov 2026	First edition. Adds My progress (D8 option 2), search, review library filters and practice shortcuts (staging).